MiSeq Loading

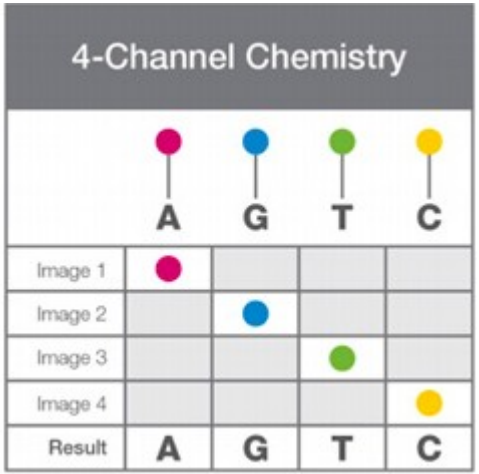


A step-by-step guide

Rapid Response – Updated Jan 2026



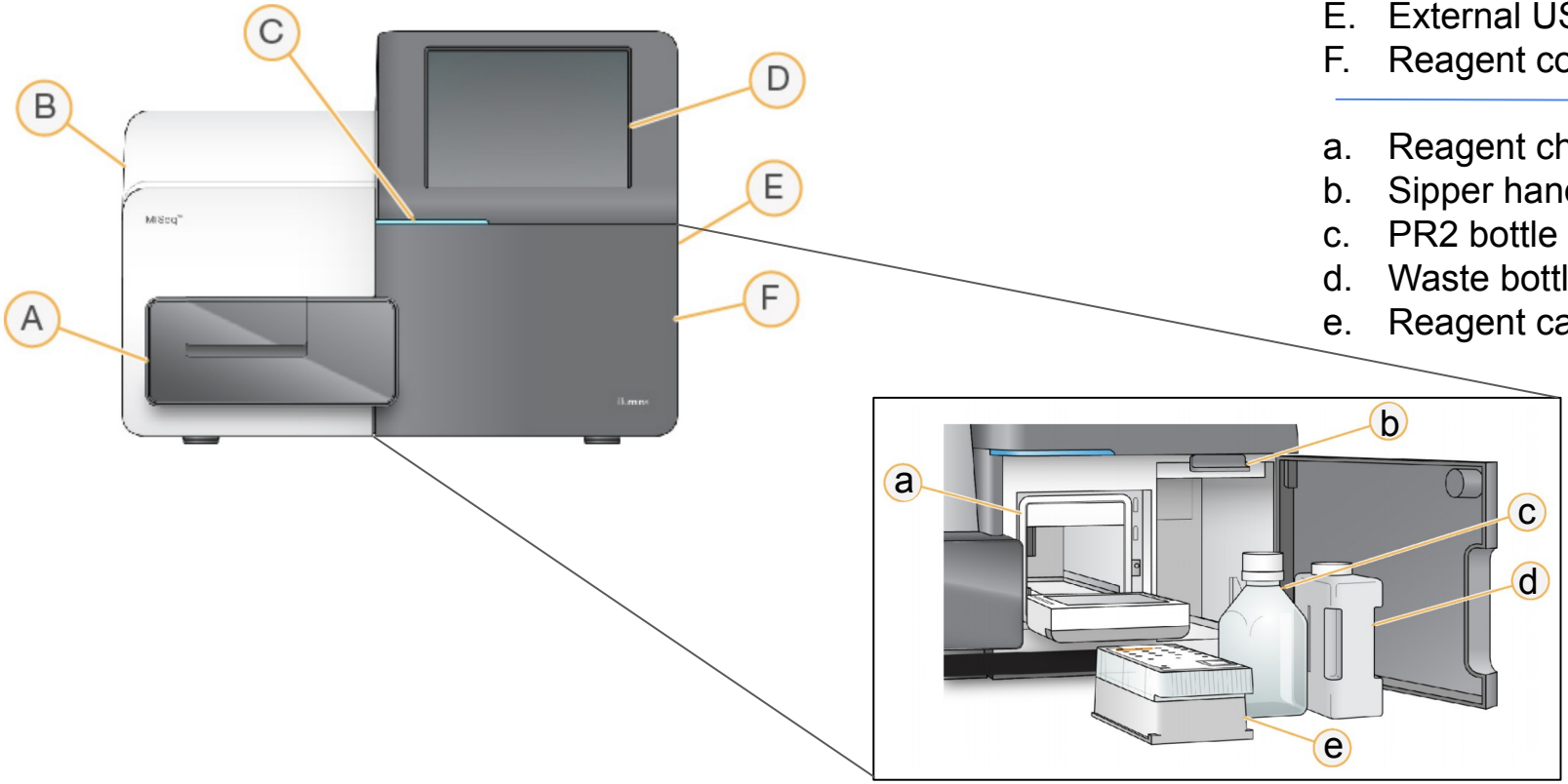
4-channel system (4 dyes)



- 1 - 24 Million reads
- 2 x 300 bp
- Max 600 cycles
- Run 4-55 hrs

- Sequencer takes four images per cycle
- **Applications:** small whole genomes (microbiome), targeted genes, 16S metagenomics
- Handles low diversity runs well

Instrument Components



A. Flow cell compartment
B. Enclosed optics compartment
C. Status bar
D. Touch screen monitor
E. External USB ports
F. Reagent compartment

---

a. Reagent chiller
b. Sipper handle
c. PR2 bottle
d. Waste bottle
e. Reagent cartridge

MiSeq Kit



Incorporation buffer (PR2)
(stored in 4°C)

Reagent cartridge
(stored at -20°C)

Hybridization buffer
(HT1)

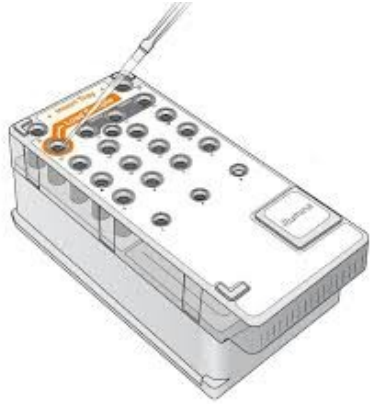
Flow cell in buffer (stored in 4°C); requires 2-3x washes with DI, followed by ethanol wipe, and drying of the flow cell with a Kimwipe

Sequencing Kits and Loading Concentrations

- MiSeq V2 (V2 - 12M, Micro- 4M, Nano-1M)
  - Available in 50, 300, 500 cycles kit
  - Our optimized loading concentration: 10pM in 600µL
  - Cluster Density Goal: 800-1000/mm^2

- MiSeq V3 (24M)
  - Available in 150, 600 cycles kit
  - Our optimized loading concentration: 12pM in 600µL
  - Cluster Density Goal: 1000-1200/mm^2

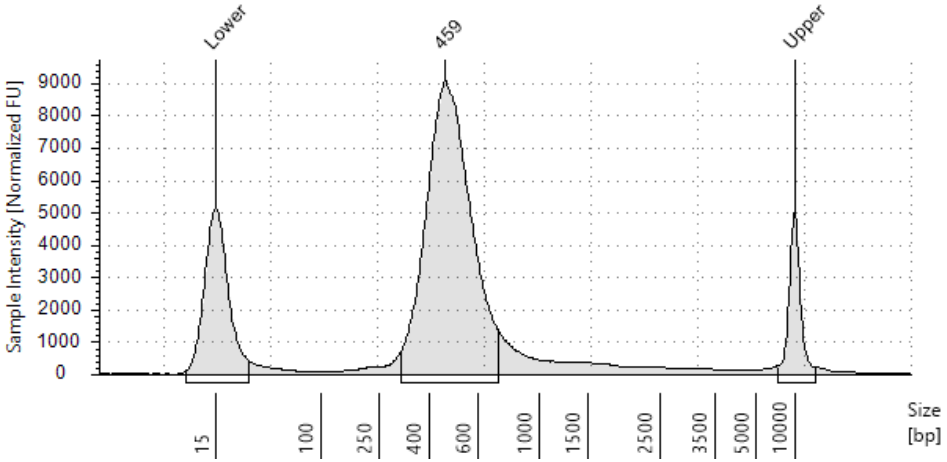
Before Loading



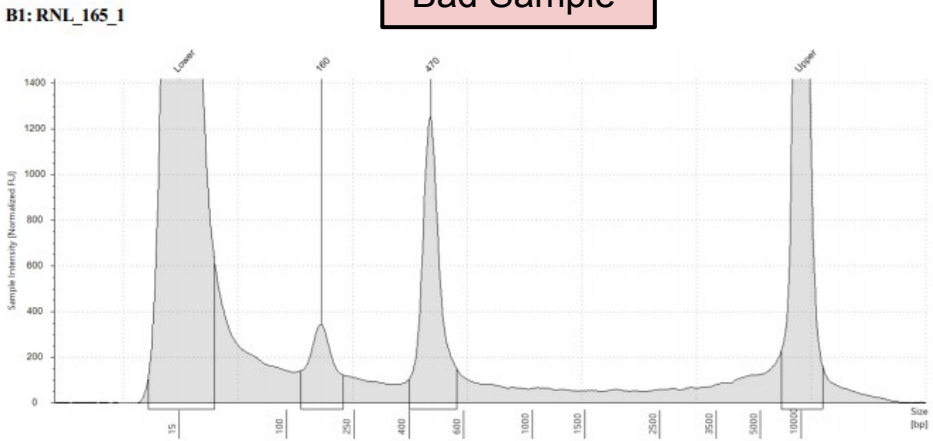
- Thaw Reagent Cartridge
  - Overnight in 4°C or water bath (room temp ~1 hr)
- Take out Flow Cell and Incorporation Buffer from 4°C
  - At least 30 minutes before loading to equilibrate to room temperature
- Invert the cartridge 5 times to mix - avoid making bubbles
- Carefully pierce foil (well #21) with a pipette tip once ready to load the diluted library into cartridge

Library Distribution

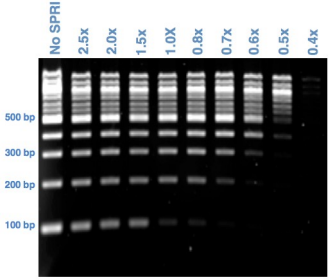
Good Sample



Bad Sample



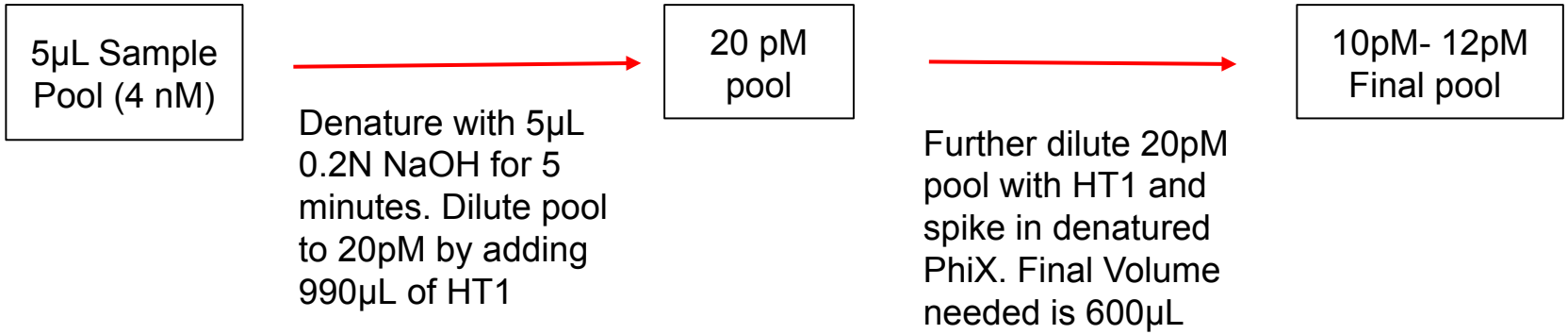
- Average fragment size should be 300-600 bp
- Fragment that is 300 base pairs long has about 150 base pairs of adapter sequences
  - Long enough to run up to PE 150
- Adapter Dimers (~140 bp) are bad because they preferentially cluster to the flowcell
  - Results in inaccurate loading
  - A small percent of dimers will take up a significant proportion of reads.



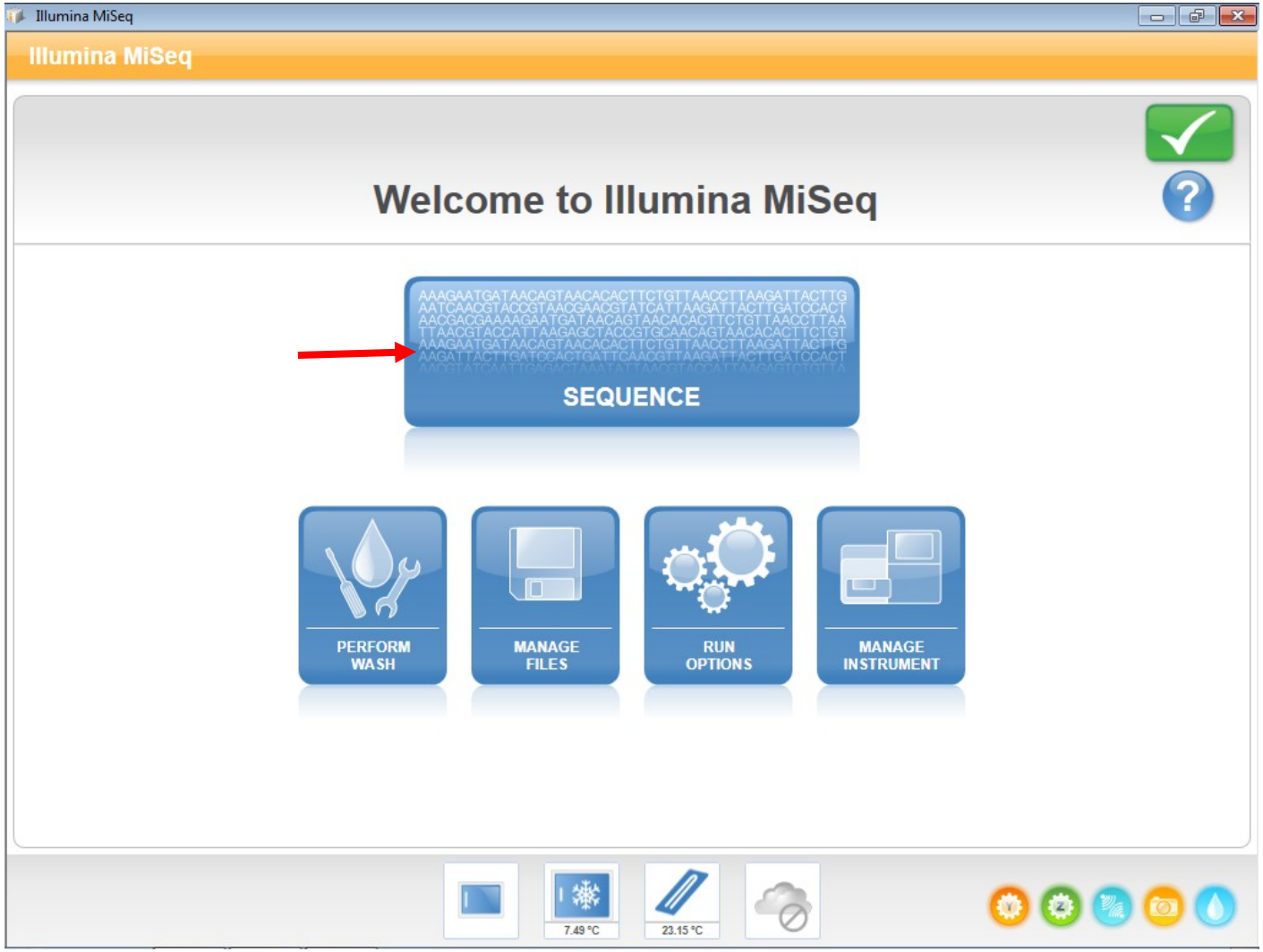
SPRI Select

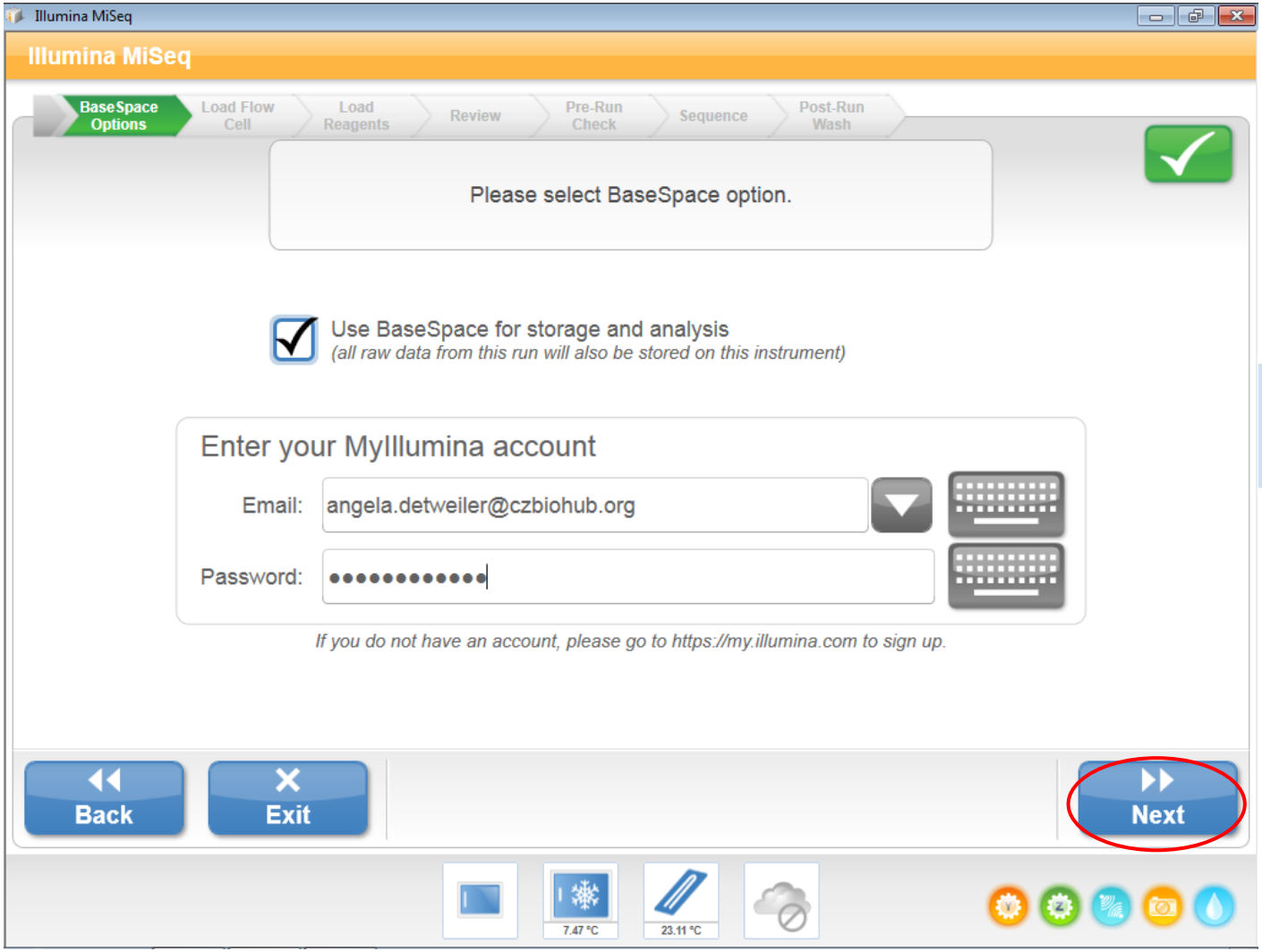
Denature and dilute library

- Quantify sample pool by bioanalyzer/tapestation and qPCR (if possible)

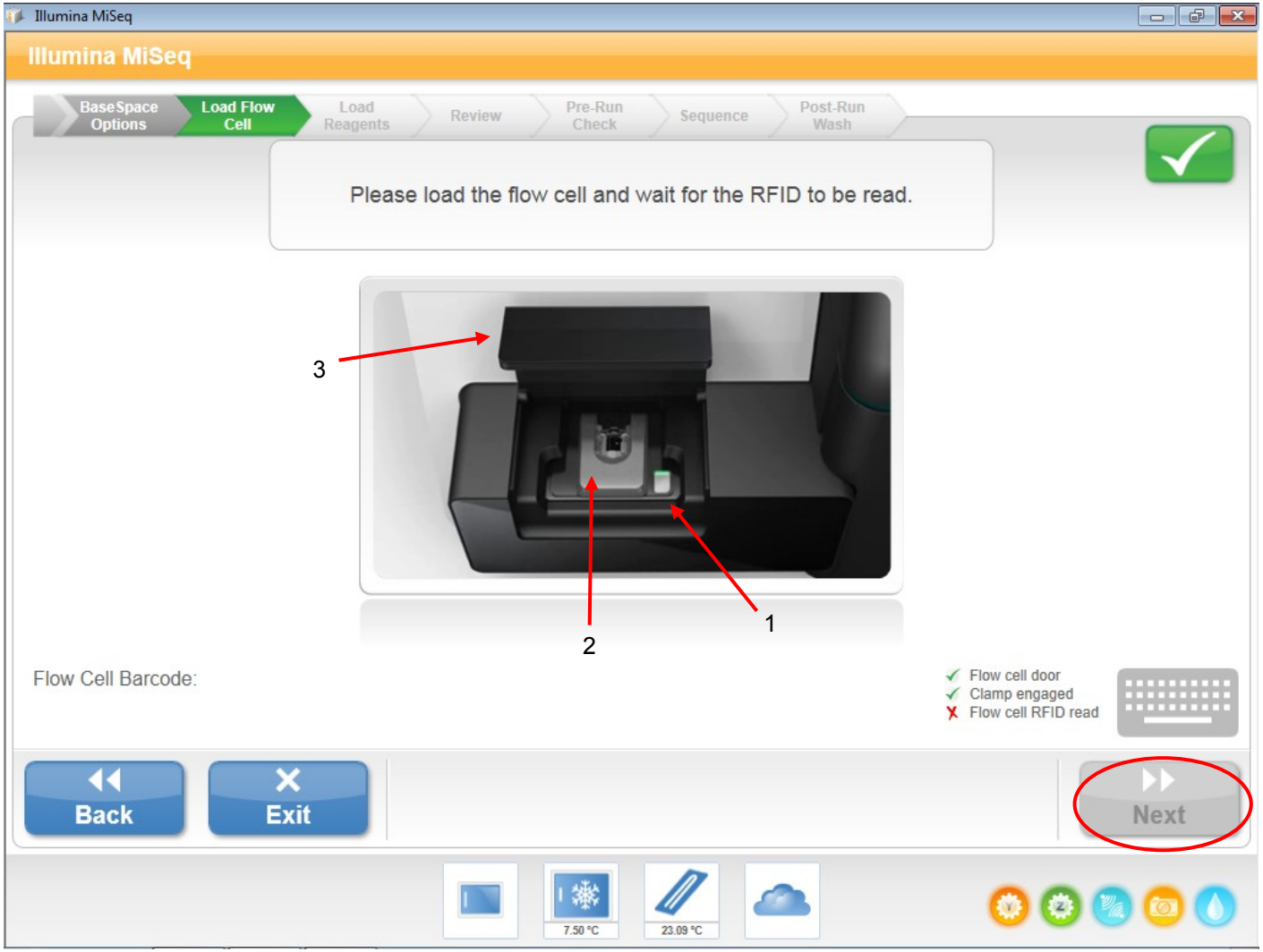


- Loading concentration can be variable between MiSeqs (keep a log)
- 1-5% PhiX recommended (15-20% for amplicons)

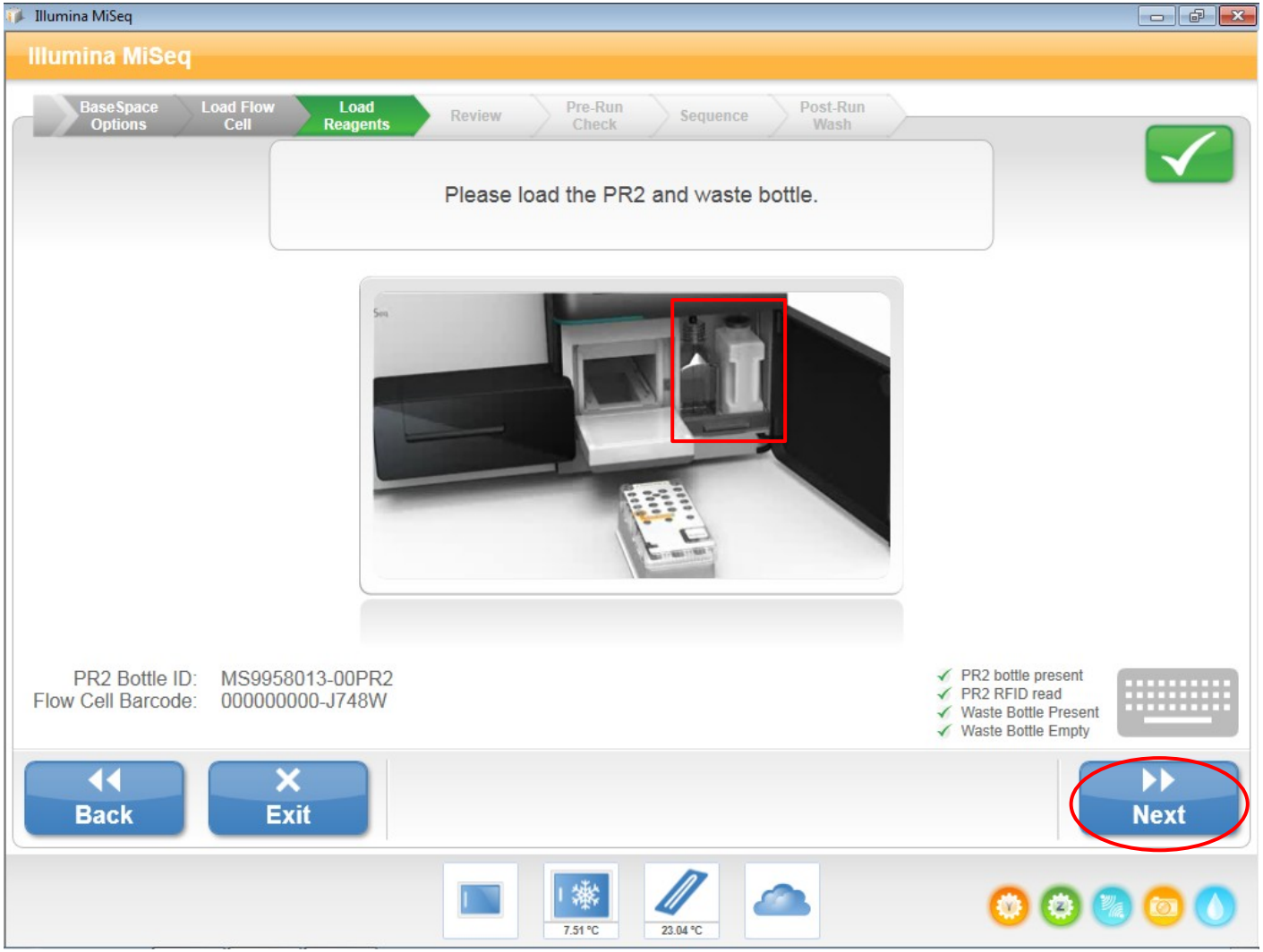

Illumina MiSeq
Illumina MiSeq
Welcome to Illumina MiSeq
SEQUENCE
PERFORM WASH
MANAGE FILES
RUN OPTIONS
MANAGE INSTRUMENT
7.49 °C
23.15 °C



Log into BaseSpace, then click Next.

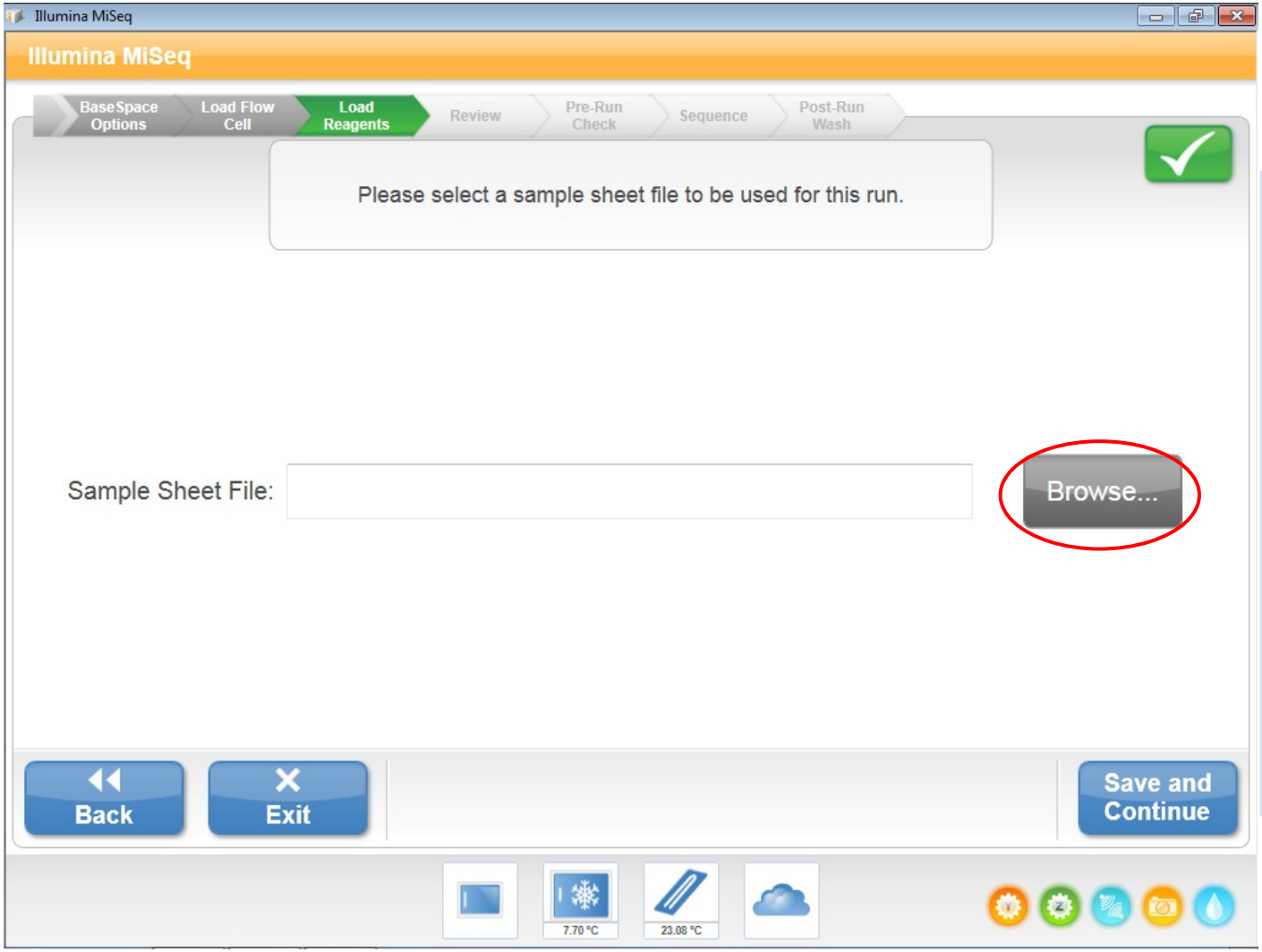


Press the white release button to open the flow cell clamp (1). Seat the flow cell and close the clamp (2). Close the flow cell compartment door (3). Once all items ✓, click “Next”.



Remove/dispose of spent reagents and return the empty waste container. Remove the wash PR2 bottle and load the new buffer. Once all items ✓, click "Next".

*Make sure that the sipper handle is down after inserting the PR2 and waste bottle. This will also come handy when troubleshooting.



The MiSeq requires that the user upload a sample sheet. At CZ Biohub, we download the sample sheet from gDrive to the Sequencer Desktop. The sample sheet name needs to be changed to the reagent cartridge ID (e.g. “MS3046361-600V3”).

Browse for the location of the file.

Samplesheet

20200723_Veronica_Escalante_CZBSeqSub03383 - Default Indices.csv — Edited

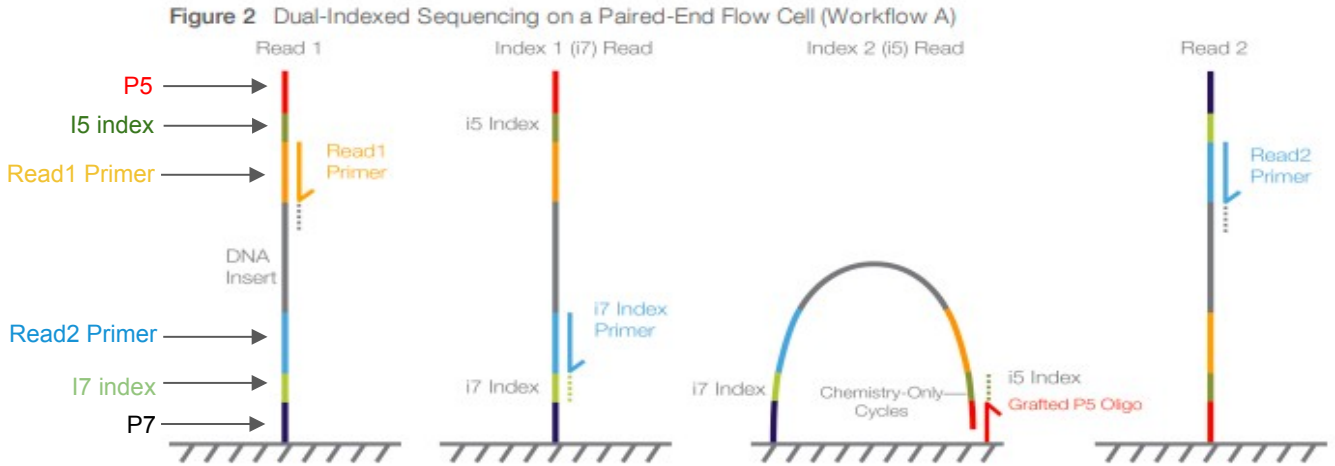
```
[Header],,,,,,,,,,,,,
Date,2020-07-23,,,,,,,,,,,,
Experiment Name, 20200723_Veronica_E,,,,,,,,
Workflow,GenerateFASTQ,,,,,,,,,,,,
Application,FASTQ Only,,,,,,,,,,,,
Chemistry,Amplicon,,,,,,,,,,,,
IEMFileVersion,4,,,,,,,,,,,,
,,,,,,,,,,,,,
[Reads],,,,,,,,,,,,,
250,,,,,,,,,,,,,
250,,,,,,,,,,,,,
,,,,,,,,,,,,,
[CZBiohubSequencing],,,,,,,,,,,,,
Submitter Name,Veronica Escalante,,,,,,,,,,,,
PI Name,Peter Turnbaugh,,,,,,,,,,,,
Task Code,4159,,,,,,,,,,,,
Submission ID,CZBSeqSub03383,,,,,,,,,,,,
Library Pool Name,SPF_SIM_RSV_VE_1,,,,,,,,,,,,
Platform,MiSeq V3,,,,,,,,,,,,
Index 1 Length,8,,,,,,,,,,,,
Index 2 Length,8,,,,,,,,,,,,
Library Number,192,,,,,,,,,,,,
Sample Sheet Status,Form  Completed   ,,,,,,,,,,,,
,,,,,,,,,,,,,
[Data],,,,,,,,,,,,,
Study_ID,Study_Description,BioSample_ID,BioSample_Description,Sample_ID,Sample_Name
,Sample_Owner,Index_ID,Index,Index2_ID,Index2,Organism,Host,Sex,Tissue_Source
,Understanding how statins shape the microbiome ,,Understanding how statins shape
```

Make sure to add “Experiment Name, XXXX,,,,,,,” in line 3 as highlighted in the sample sheet example here. This will name the run on BaseSpace.

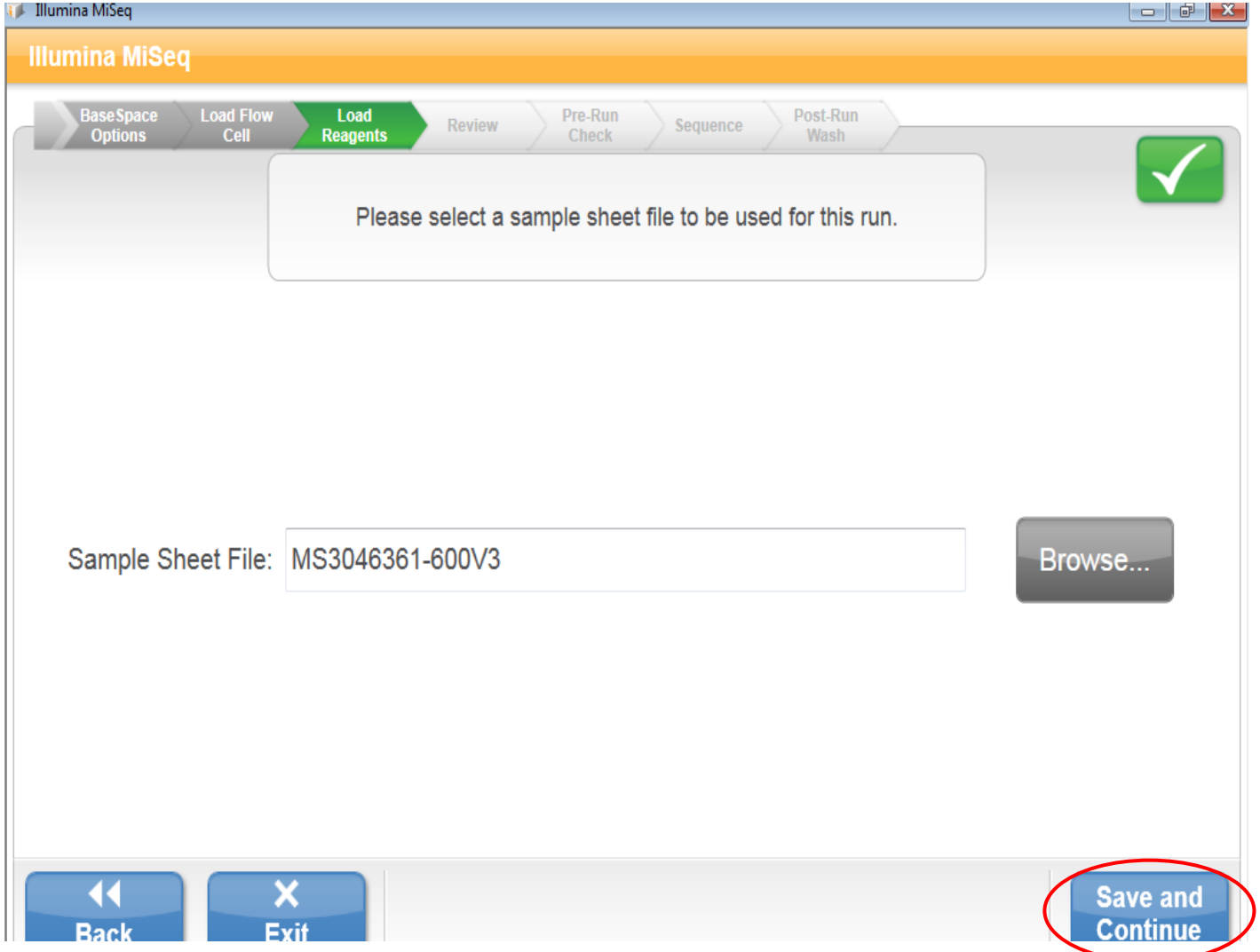
Sample sheet example

Index Reads

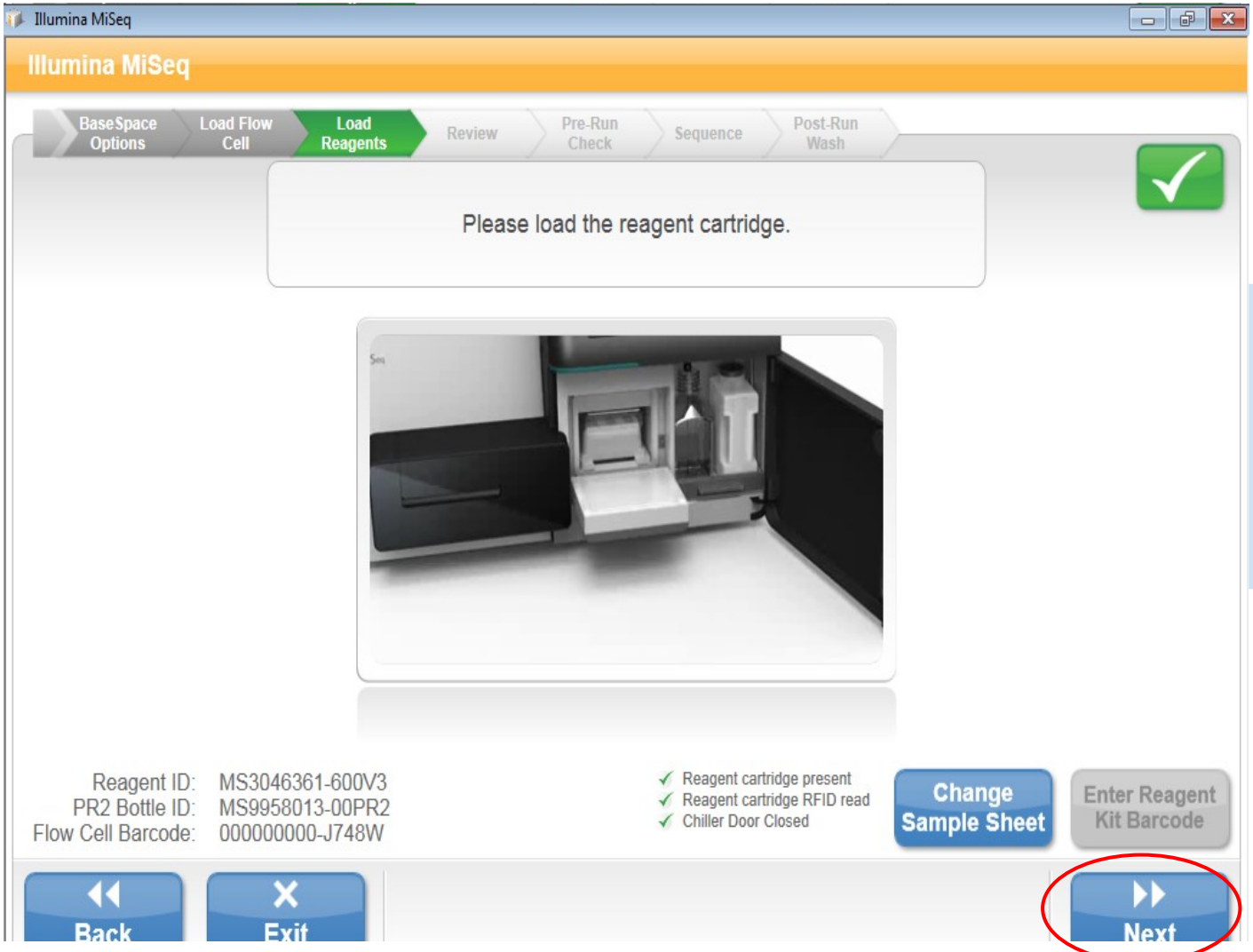
- When preparing a sample sheet for a MiSeq run-
  - I7 index (Index 1) sequence is the reverse complement of the ordered sequence
  - I5 index sequence (Index 2) is the sequence that you ordered



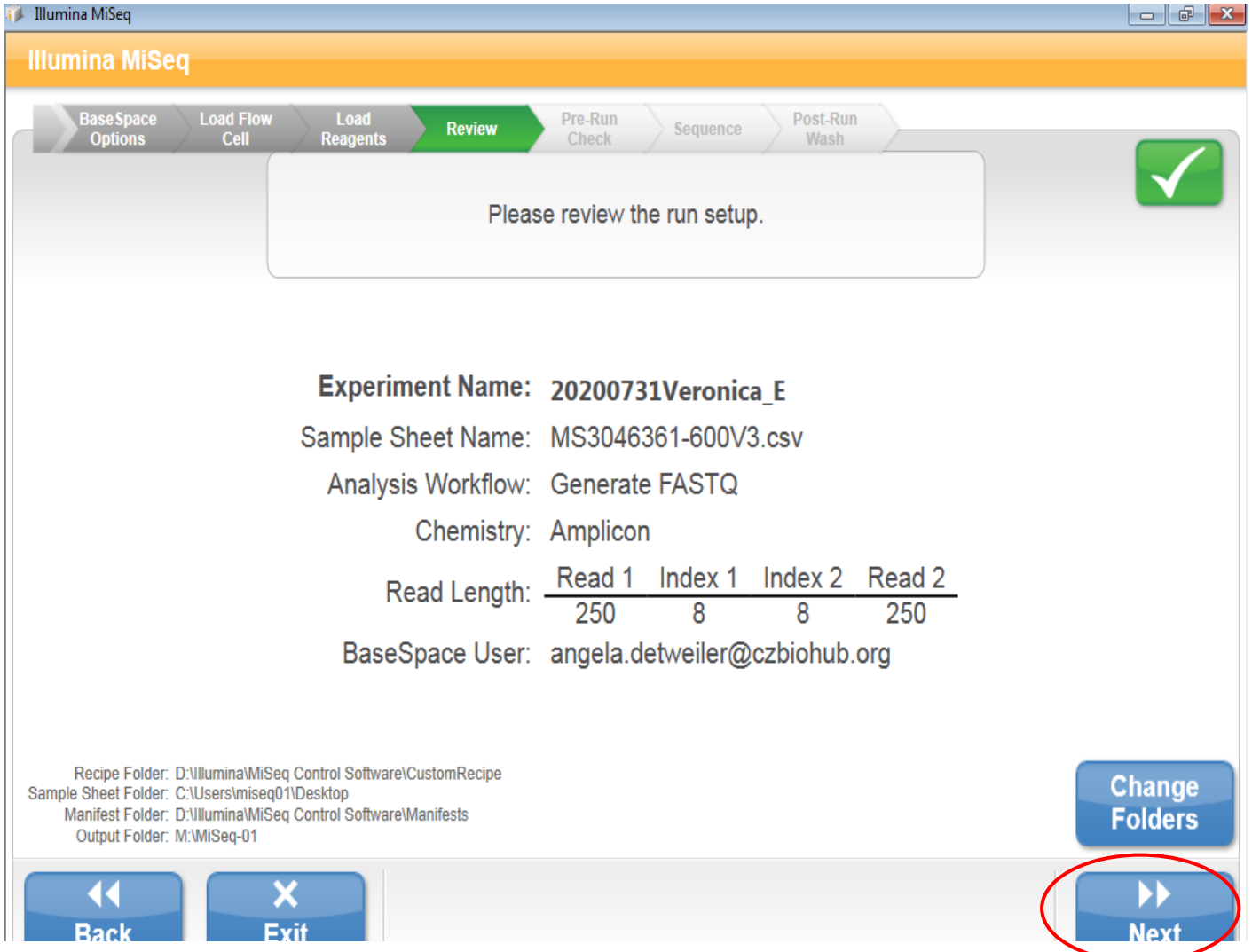
Illumina Sample sheet documentation



Once the sample sheet is selected, click “Save and Continue”



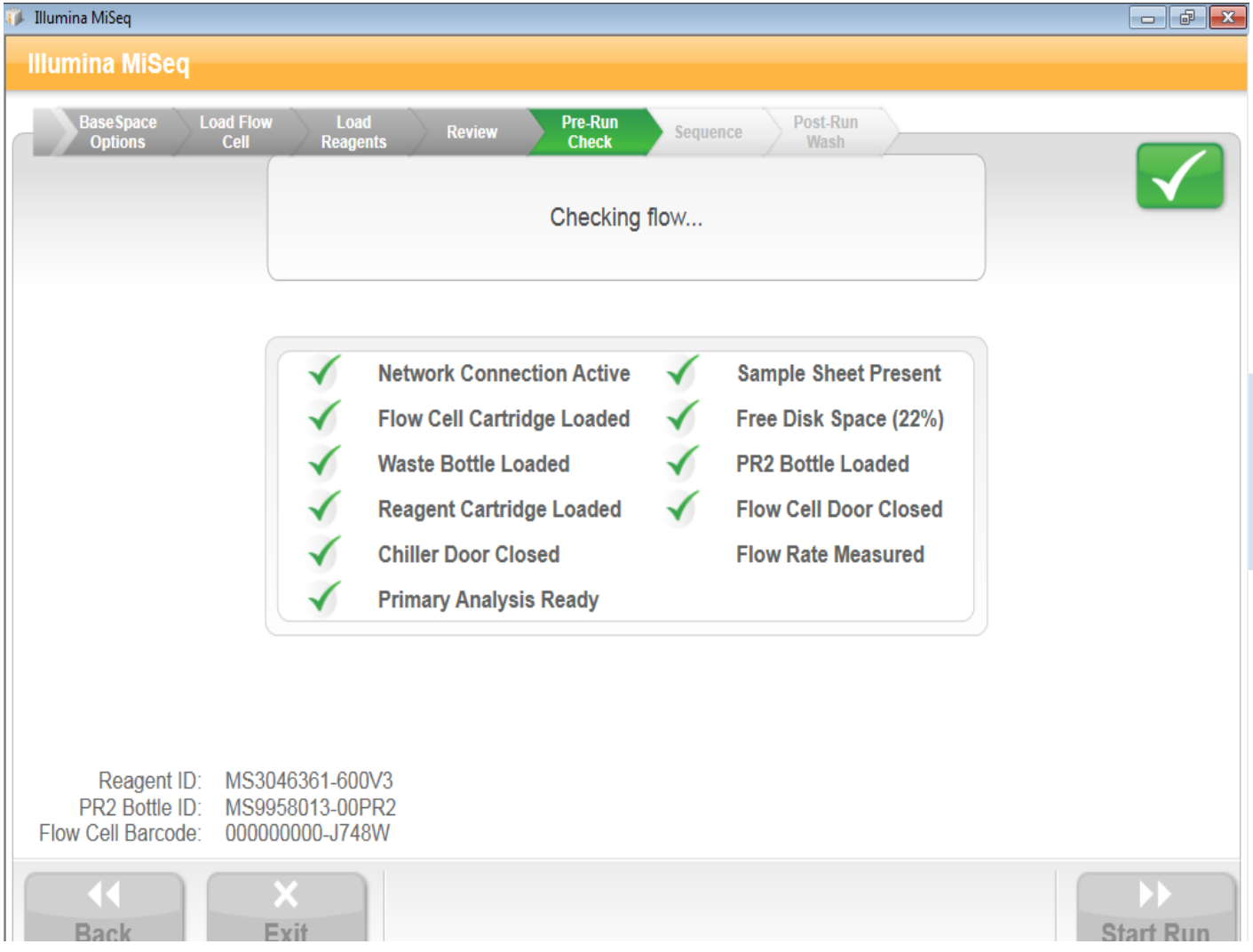
Finally remove the the wash cartridge, and slide in the new reagent cartridge. Close the reagent cartridge doors. Once all items ✓, click “Next”.



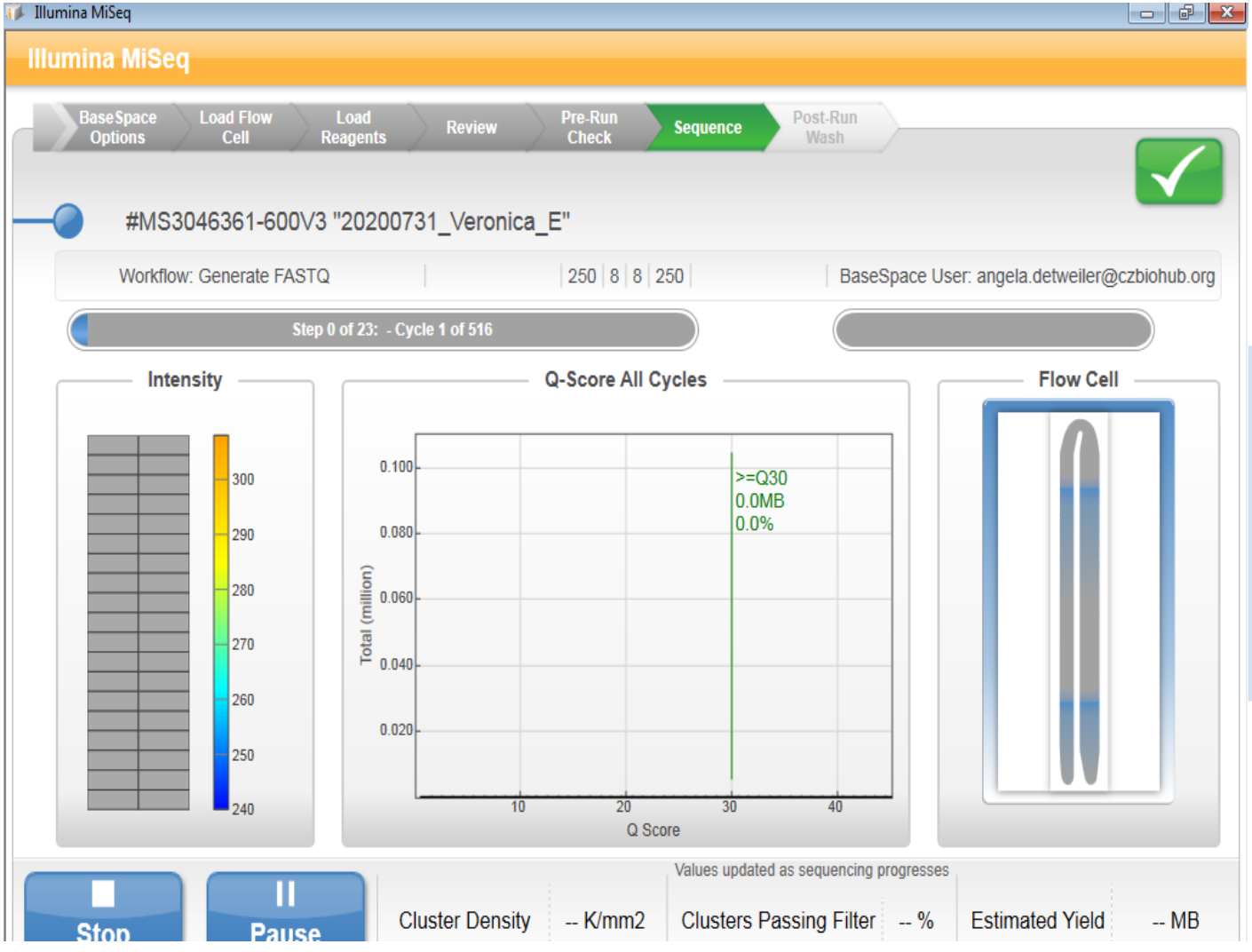
Review the run setup information pulled from the uploaded sample sheet.

If the Read or Index length needs to be updated, go back to the sample sheet, make changes and re-upload the sample sheet.

Click “Next”.



The pre-run check will take a few minutes. Once all items ✓, click “Start Run”.



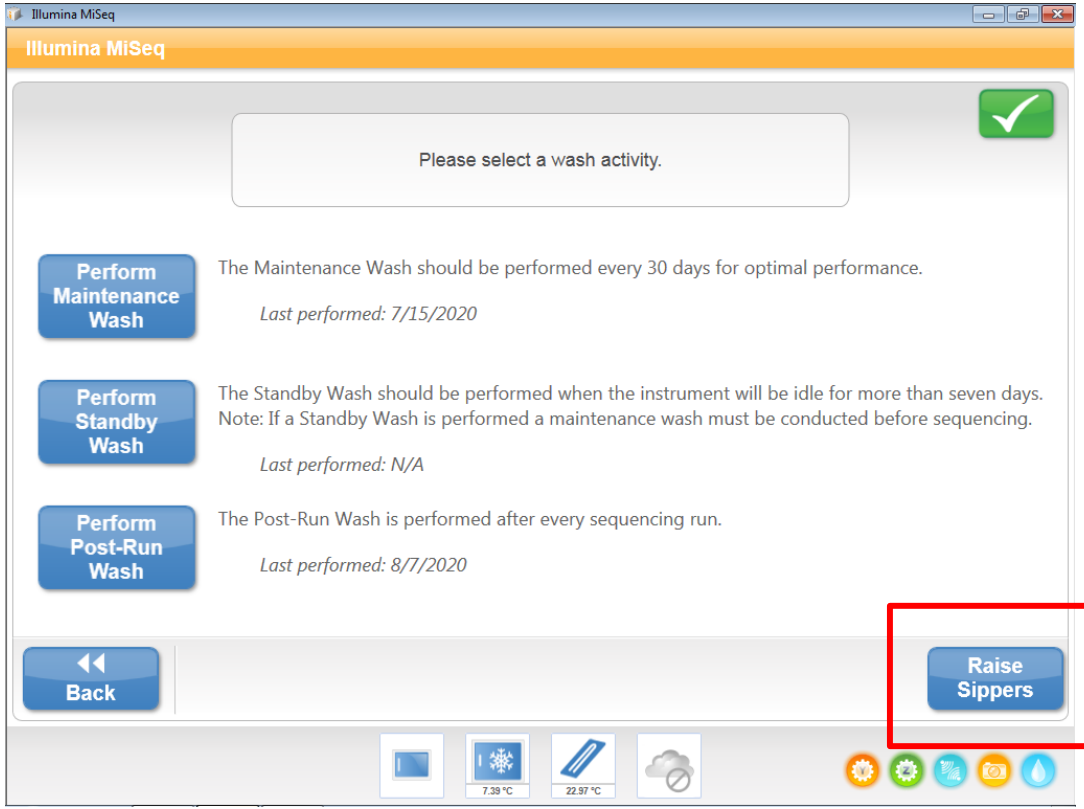
Cluster density and PF% metrics will become available after 20 and 25 cycles, respectively.

MiSeq runs can last from 4-56 hrs depending on the kit.

Perform Wash


Illumina MiSeq
Illumina MiSeq
Welcome to Illumina MiSeq
SEQUENCE
PERFORM WASH
MANAGE FILES
RUN OPTIONS
MANAGE INSTRUMENT
7.49 °C
23.15 °C

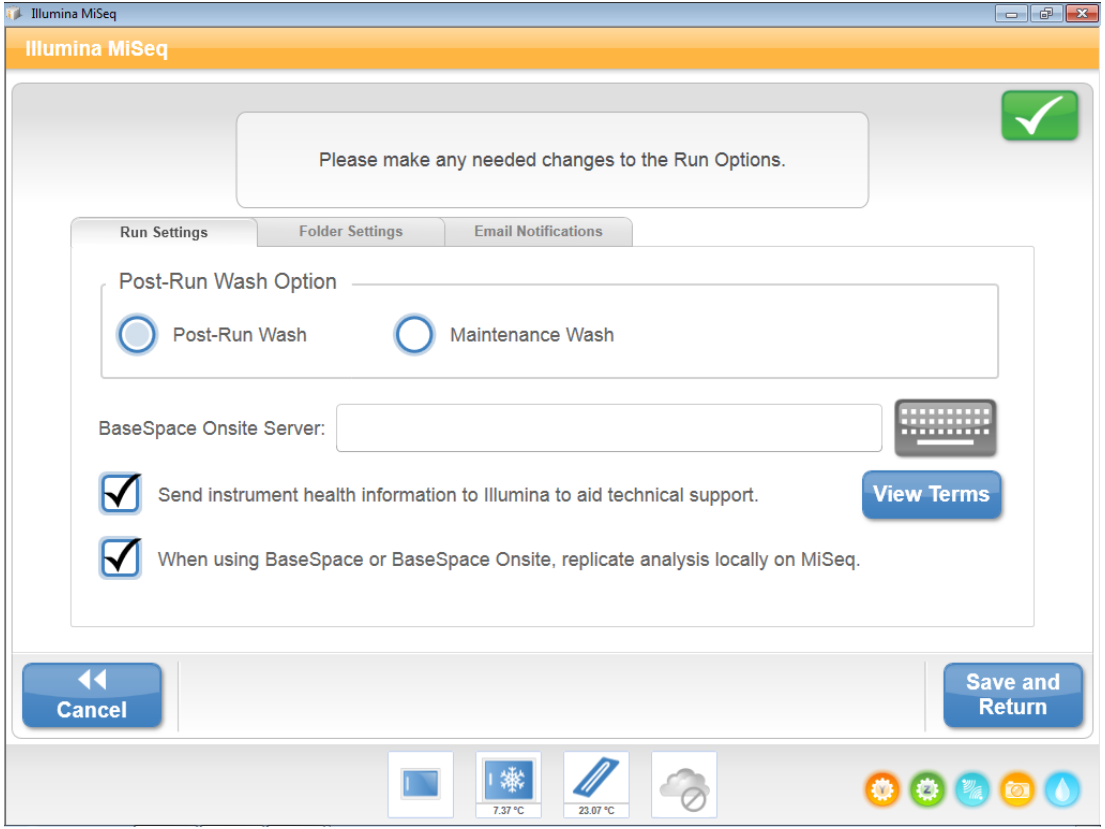
Wash Types



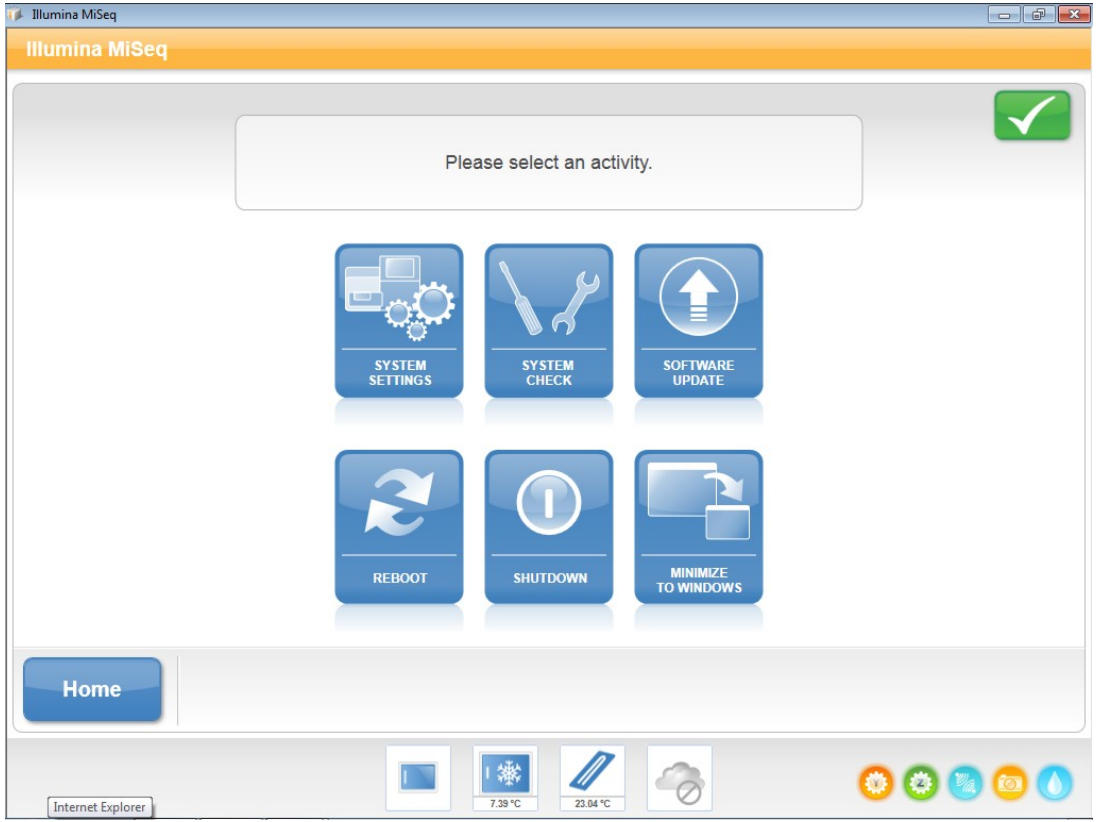
- **Maintenance Wash:** Once every month (~90mins)
  - A series of three wash steps with 0.5% Tween20
- **Stand by Wash:** Only if machine will be idle for for more than a week.
  - A series for two washes with 0.5% Tween20
- **Post-Run Wash:** Mandatory after every run with 0.5% Tween20

When troubleshooting, you may need to raise the sippers manually

Run Options


Illumina MiSeq
Illumina MiSeq
Please make any needed changes to the Run Options.
Run Settings
Folder Settings
Email Notifications
Post-Run Wash Option
Post-Run Wash
Maintenance Wash
BaseSpace Onsite Server:
Send instrument health information to Illumina to aid technical support.
View Terms
When using BaseSpace or BaseSpace Onsite, replicate analysis locally on MiSeq.
Cancel
Save and Return
7.37 °C
23.07 °C

Manage Instrument



Next: 08-I_Sequencing Run QC for Illumina – Slides & Activity